

Experiment No: 07

Date : _____

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Aim : To Draw Different Types of Engineering Curves.

Theory :

Engineering Curves :

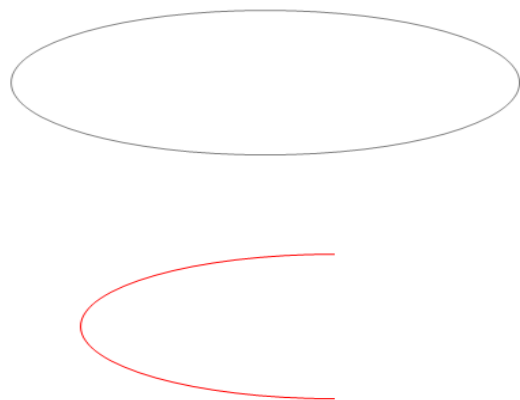
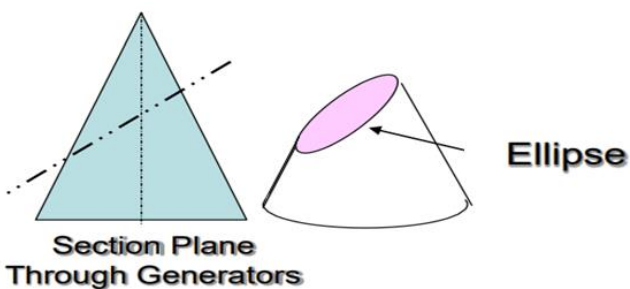
The curves obtained by the intersection of a cone by cutting plane in different position is called as engineering curves or conics.

Engineering Curves :

- Ellipse
- Parabola
- Hyperbola
- Cycloid
- Epicycloid
- Hypocycloid
- Involute

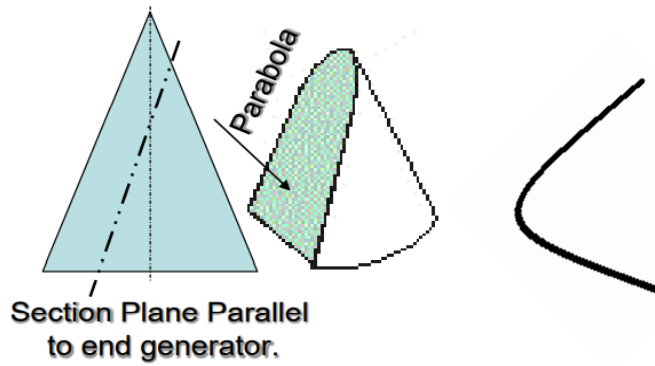
Ellipse :

A regular oval shape , traced by a point moving in a plane so that sum of distances from two other point is same.



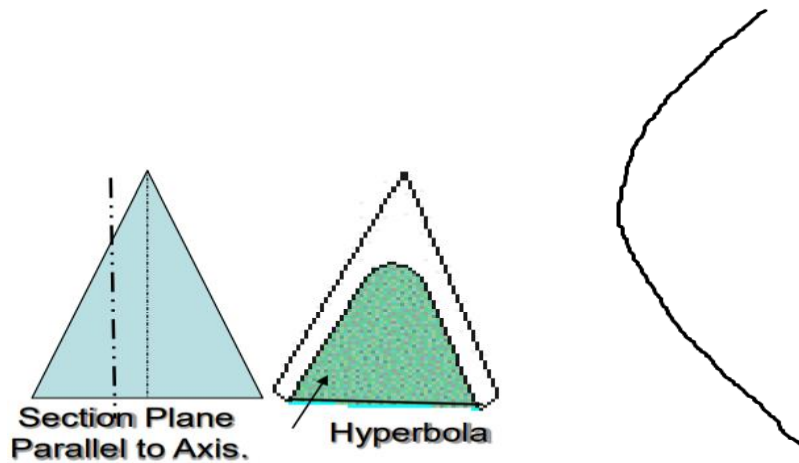
Parabola :

It is a symmetrical plane curved formed by the intersection of a cone with a plane parallel to its side. The path of a projectile under the influence of gravity follows a curve of this shape.



Hyperbola:

A symmetrical open curve formed by the intersection of circular cone with a plane at a smaller angle with its axis than the side of cone



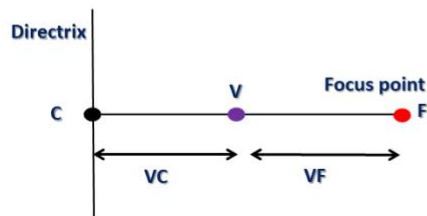
Methods to draw engineering curves:

1. Directrix circle method
2. Arc of circle method
3. Rectangle method
4. Concentric circle method.

1. Directrix circle method :

- **Focus :** It is a fixed point
- **Directrix :** It is a fixed straight Line
- **Axis :** It is a fixed straight Line at right angles to directrix and passing through the focus
- **Vertex :** It is a point where conic intersects the axis.
- **Eccentricity :** It is a ratio of the distance of moving point from the focus to its distance from the directrix. It is denoted by 'e'

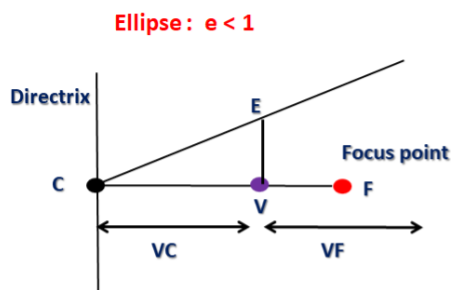
$$\text{Eccentricity (e)} = \frac{\text{Distance of the point from the focus (VF)}}{\text{Distance of the point from the directrix (VC)}}$$



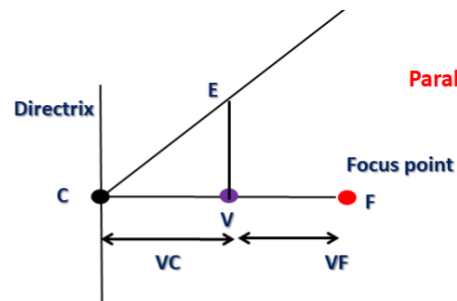
Ellipse: $e < 1$

Hyperbola: $e > 1$

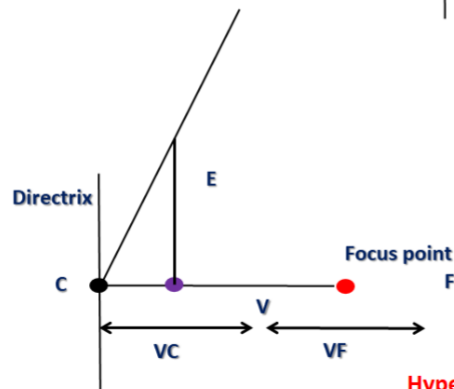
Parabola: $e = 1$



Ellipse: $e < 1$



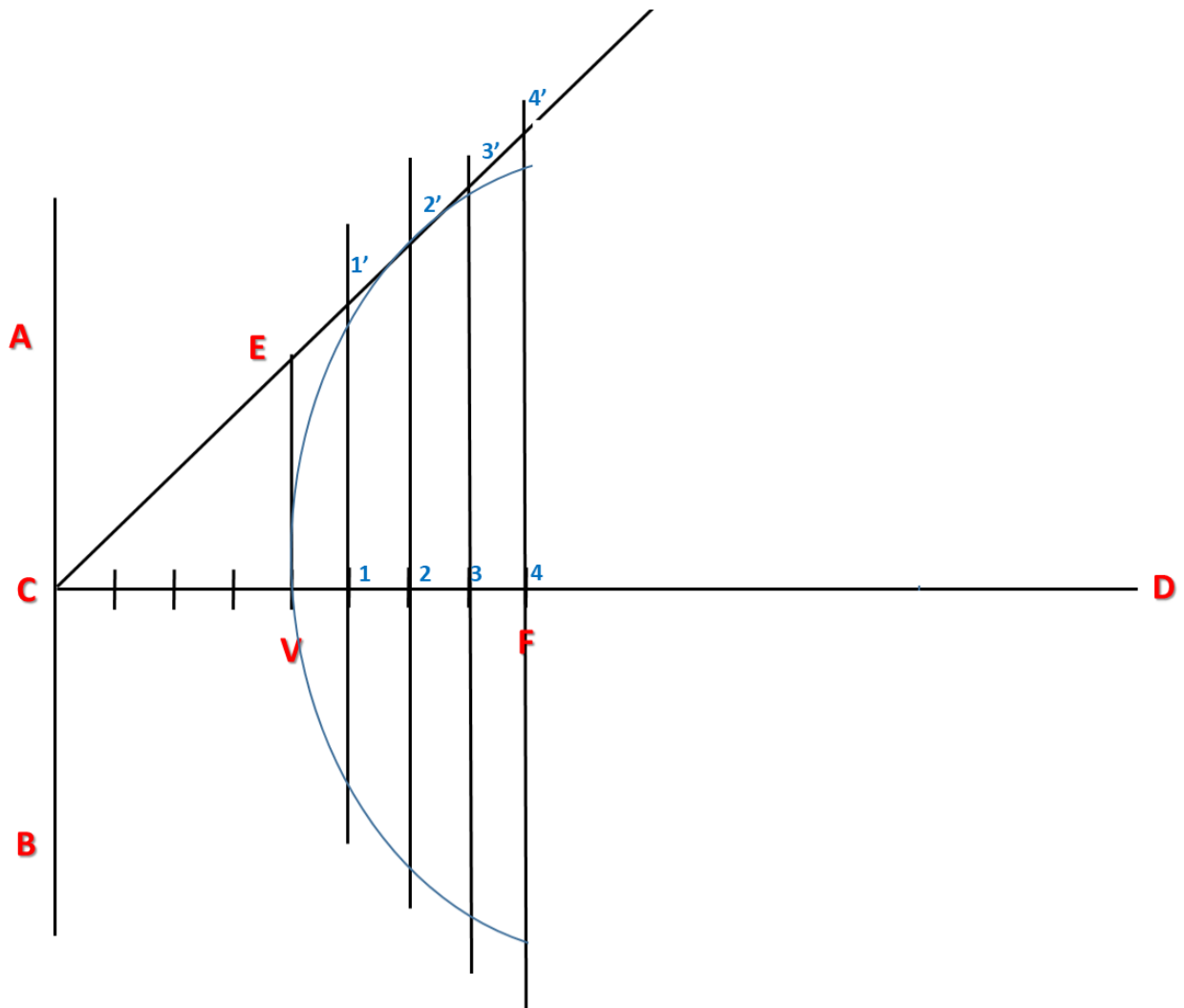
Parabola: $e = 1$



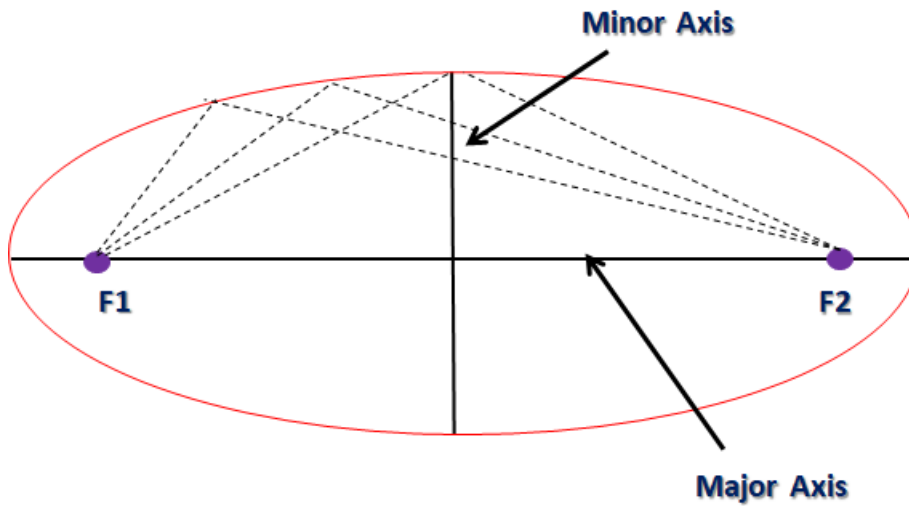
Hyperbola: $e > 1$

Problem 1 : A fixed point F is 80 mm from a fixed straight line AB. Draw parabola by using directrix focus method.

Given Data : Focus Point F = 80 mm , eccentricity = $\frac{4}{4}$



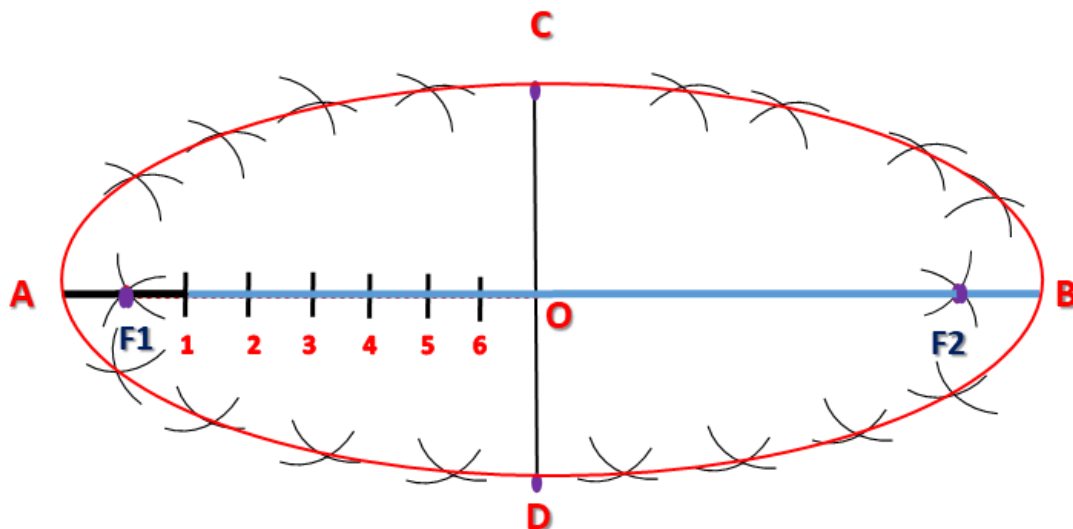
2. Arc of circle method :



Problem :

The major axis of an ellipse is 150 mm long and minor axis is 100 mm long. Find Foci and draw an ellipse by 'Arc of circle method'

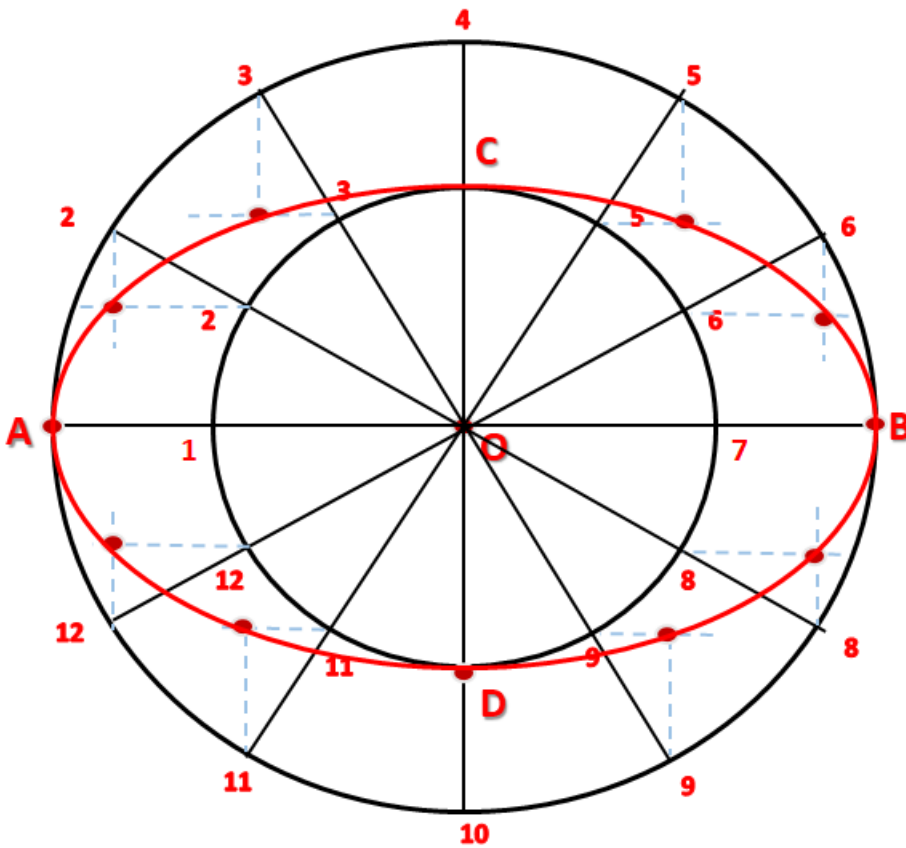
Given data : Major Axis AB = 150 mm , Minor Axis CD = 100 mm



3. Concentric circle method :

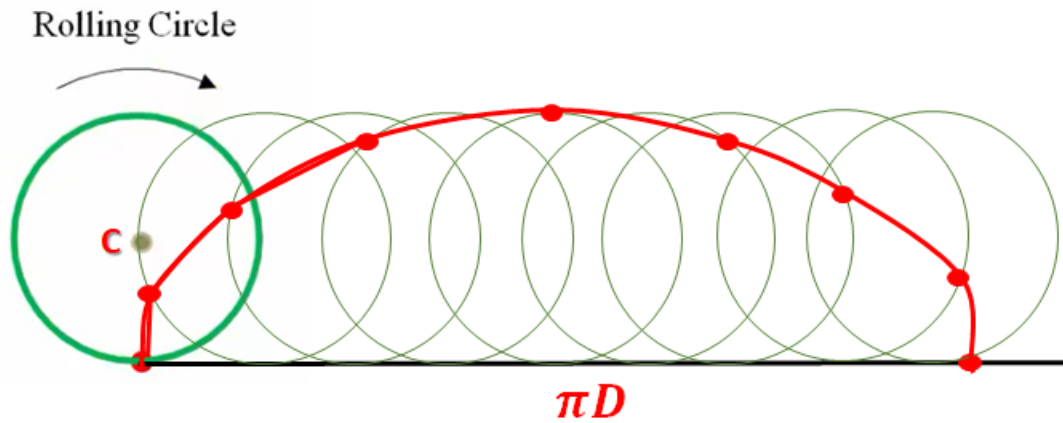
The major axis of an ellipse is 100 mm long and minor axis is 70 mm long. Draw an ellipse by 'Concentric Circle Method'

Given data : Major Axis AB = 100 mm , Minor Axis CD = 70 mm



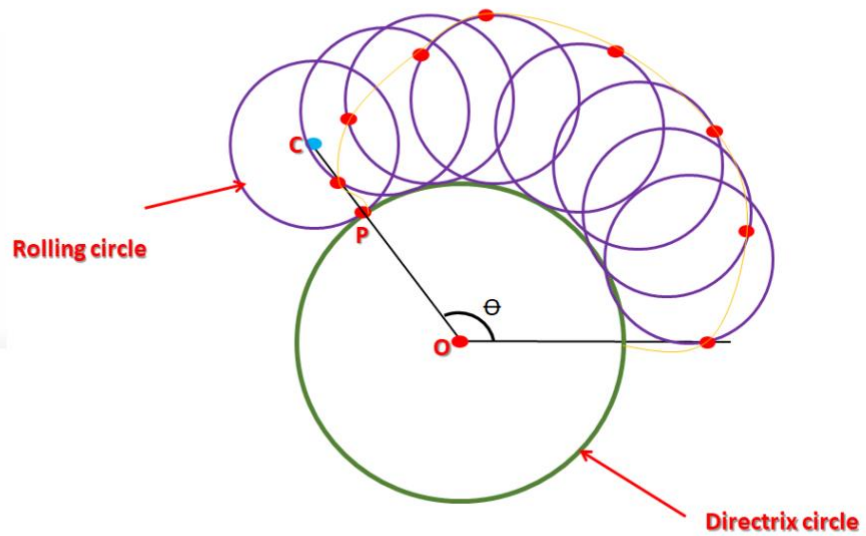
Cycloid :

A Cycloid is the curve generated by a point "P" on the circumference of a circle which moves along a straight line without slipping.



Epicycloid

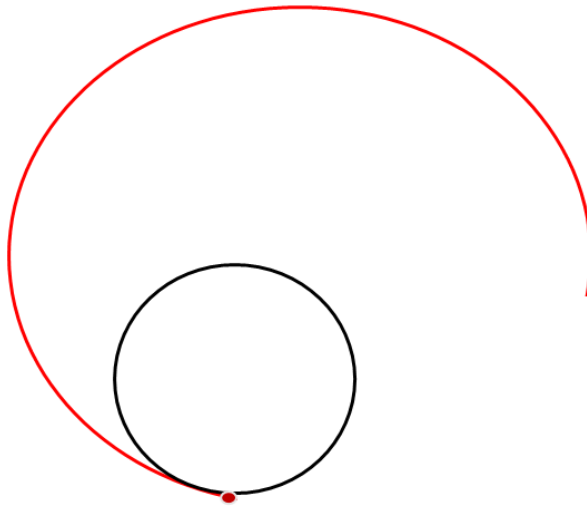
An *Epicycloid* is the curve generated by a point "P" on the circumference of a circle (rolling) which rolls outside a circle (directrix).



$$\theta = \text{radius of rolling circle (r)} / \text{radius of directrix circle (R)}$$

Involute

Involute is a single curved line traced out by an end of a string when unwound itself from a circle.



Conclusion :